The Starfish:
a Giant Step Toward Origins of Life

"If I followed correctly, the circular RNA sequence obeys the Fibonacci rule. If we extrapolate, we can think that Life was formed (or was created, according to our religion) from this RNA according to a mathematical principle? "
(Luc Montagnier)

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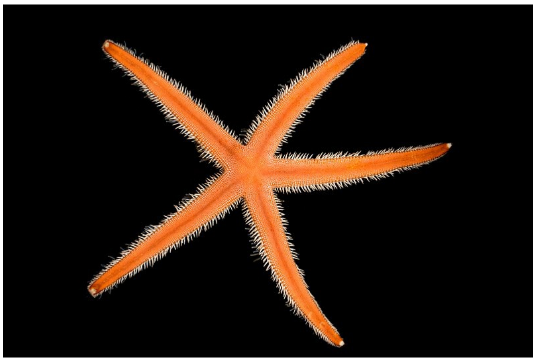
ABSTRACT :

Here we study the circular RNA sequence of 22 bases named ALpha, which is supposed to be a candidate for the emergence of life on earth. We demonstrate how this sequence is structured by Fibonacci proportions, then we demonstrate that, among the millions of known sequences, it is in the starphish genome that we find this sequence with the highest homology, thus uniting the scales of RNA and morphogenesis since the geometry of the starfish is also structured by the Golden Ratio.

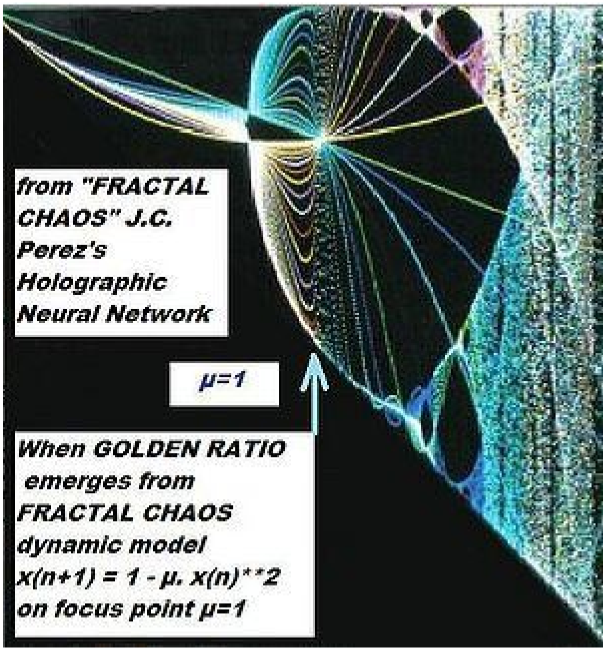
-I- the Starfish:

The Starfish has always fascinated people because of its great aesthetics. Beauty inherited from its geometric link with the 5-pointed star and the pentagon in which the ratio of the diagonal to the side is the famous Golden Ratio Phi = 1.618033. (Perez JC. Codon populations in single-stranded whole human genome DNA Are fractal and fine-tuned by the Golden Ratio 1.618. Interdiscip Sci. 2010 Sep;2(3):228-40. doi: 10.1007/s12539-010-0022-0. Epub 2010 Jul 25. Erratum in: Interdiscip Sci. 2010 Dec;2(4):373. PMID: 20658335 https://pubmed.ncbi.nlm.nih.gov/20658335/). Which corresponds for the starfish to the ratio of the distances between (one branch and the next of its neighbor) and (one branch and its neighbor). Or if we number the branches from 1 to 5, (distance 1 and 3) / distance 1 and 2). It was not until 2023 that the entire genome of the beautiful and intriguing STARFISH was sequenced (https://www.ncbi.nlm.nih.gov/nucleotide/OX465102.1?report=genbank&log$=nuclalign&blast_rank=1&RID=XN5H46XM016).It is this recent discovery of the genomic sequence of the starfish which will provide new insight into the scientific quest for the origins of terrestrial life...



The Starfish

-II- Is there an original RNA?



When Golden Ratio emerges from Fractal Chaos

As in the fractal universe, where everything is self-similar at all scales, let us now travel towards the infinitely small universe of RNA which is said to be at the origins of life on earth or rather in the seas. .. Like the star of the sea... From 2007 in A possible circular RNA at the origin of life (Demongeot, J. (2007). A possible circular RNA at the origin of life. Journal of Theoretical Biology. https://www.academia.edu/76445494/A_possible_circular_RNA_at_the_origin_of_life?email_work_card=title)

Demongeot and Moreira, Then In (Demongeot, J.; Norris, V. Emergence of a “Cyclosome” in a Primitive Network Capable of Building “Infinite” Proteins. Life 2019, 9, 51. https://doi.org/10.3390/life9020051), Demongeot and Norris argue for the existence of an RNA sequence, called the AL (for ALpha) sequence, which may have played a role at the origin of life;

Here is this “ring” sequence that they called “ALpha”:

AUGGUACUGCCAUUCAAGAUGA

-III- Fibonacci proportions structure the original ALpha RNA sequence:

In (Perez, J.C. (2021). SARS-COV2 VARIANTS AND VACCINES MRNA SPIKES FIBONACCI NUMERICAL UA/CG METASTRUCTURES. International Journal of Research -GRANTHAALAYAH, 9(6), 349-396. https://doi.org/10.29121/granthaalayah.v9.i6.2021.4040) with RIP Luc Montagnier, we demonstrated how this strange circular RNA "ALpha ring" sequence, 22 nucleotides long (AUGGUACUGCCAUUCAAGAUGA) obeyed the Fibonacci proportions for the UA/CG populations throughout the ALpha circular ring.

We also note 2x6 = 12 bases in primers of palindromes or mirror series:

AUGGUA mirror and UCAAGA quasi palindrome.

Let's organize this circular sequence into codon triplets and gradually shift it codon triplet by codon triplet.

```
=> Aug GUA CUG CCA UUC AAG AUG A 13AU & 8CG
Au GGU ACU GCC AUU CAA G => Aug A 13AU & 8CG
Au GGU ACU GCC AUU C => AAG Aug A 13AU & 8CG
Au GGU ACU GCC A => UUC AAG Aug A 13AU & 8CG
Au GGU ACU G => CCA UUC AAG Aug A 13AU & 8CG
Au GGU A => CUG CCA UUC AAG Aug A 13AU & 8CG
Au G => GUA CUG CCA UUC AAG Aug A 13AU & 8CG
```

This sequence is entirely structured by proportions of nucleotides obeying the proportions of the Fibonacci sequences (13AU and 8CG). Let's drag this sequence gradually one nucleotide at a time:

AUGGUACUGCCAUUCAAGAUGA

AUGGUACUGCCAUUCAAGAUGA
AAUGGUACUGCCAUUCAAGAUG
GAAUGGUACUGCCAUUCAAGAU
UGAAUGGUACUGCCAUUCAAGA
AUGAAUGGUACUGCCAUUCAAG
GAUGAAUGGUACUGCCAUUCAA
... /...

GAUGAAUGGUA CUGCCAUUCAA

GAUGAAUGGUA
AACUUACCGUC

Weak matches (UA. vs. CG)

GAUGAAUGGUA
! ! !! ! ! !! !!
AACUUACCGUC

Strong matches.
GAUGAAUGGUA
! !! ! ! !! !
AACUUACCGUC

So this circular RNA can fold into a sort of “hairpin”.
We then see that, less constrained than a Formal correspondence, CG or UA, the more flexible form of UA/CG correspondences allowed by the Fibonacci proportions also constitutes a way of locating palindromes.

There is therefore a certain potential for PALINDROME
There is therefore a certain potential for PALINDROME type symmetry where circular RNA when folding highlights UA and CG base pairs, as in DNA.

Particularly, this hyper constrainst circular 22nt sequence codes for the twenty amino acids + codon stop + only one redundant amino acid (MET).

Seen from the point of view of the autopoiesis Francisco Varela theory (ref Varela), autonomy of Indoor vs. Outdoor systems, these results could be interpreted as the rest of the loop of 21 UACG (outdoor) "seen" from a base C or G (indoor). So, the perceived signal is a kind of Fibonacci resonance ... By virtue of Francisco Varela's theory of autopoiesis (Varela § Maturana, 1980) that we applied to Artificial Intelligence in the 1980s by creating the “fractal chaos” artificial neural network (Perez 1988). Thus, the Fibonacci numbers, therefore the optimal proportion of the Golden Tatio would perhaps have already been present from the first moments of life on earth, a life for which they would have served as a "matrix" ...

-IV- Among the millions of sequences and genomes sequenced, which one is closest to ALpha RNA?

We searched using BLASTn for the closest homologies to this famous original ALpha candidate sequence.

Here are the 3 closest sequences:

The 3 sequences closest to the original RNA

AUGGUACUGCCAUUCAAGAUGA

Homologies in descending order:

1/ ==> Starfish 21/22

The Starfish

2/ ==> Protonemura montana insect 20/22

Protonemura montana insect

3/ ==> Armillaria ostoyae mushroom 19/22

Armillaria ostoyae mushroom

Details:

Luidia sarsii genome assembly, chromosome: 2

Sequence ID: OX465102.1Length: 31917233Number of Matches: 1

Range 1: 25087392 to 25087412GenBankGraphicsNext MatchPrevious Match

Alignment statistics for match #1

Score	Expect	Identities	Gaps	Strand
42.1 bits(21)	0.65	21/21(100%)	0/21(0%)	Plus/Minus

```
Query  1         ATGGTACTGCCATTCAAGATG  21
                 |||||||||||||||||||||
Sbjct  25087412  ATGGTACTGCCATTCAAGATG  25087392
```

DownloadGenBankGraphicsNextPreviousDescriptions

Protonemura montana genome assembly, chromosome: 9

Sequence ID: OX387611.1Length: 13083817Number of Matches: 1

Range 1: 12093441 to 12093460GenBankGraphicsNext MatchPrevious Match

Alignment statistics for match #1

Score	Expect	Identities	Gaps	Strand
40.1 bits(20)	2.6	20/20(100%)	0/20(0%)	Plus/Plus

```
Query  2         TGGTACTGCCATTCAAGATG  21
                 ||||||||||||||||||||
Sbjct  12093441  TGGTACTGCCATTCAAGATG  12093460
```

DownloadGenBankGraphicsNextPreviousDescriptions

Armillaria ostoyae strain C18/9 genome assembly, chromosome: LG4

Sequence ID: LR732078.1Length: 5066647Number of Matches: 1

Range 1: 1469717 to 1469735GenBankGraphicsNext MatchPrevious Match

Alignment statistics for match #1

Score	Expect	Identities	Gaps	Strand
38.2 bits(19)	10	19/19(100%)	0/19(0%)	Plus/Plus

```
Query  1        ATGGTACTGCCATTCAAGA  19
                |||||||||||||||||||
Sbjct  1469717  ATGGTACTGCCATTCAAGA  1469735
```

-V- Conclusion:

Three conclusions emerge from this interdisciplinary study:
On the one hand, even if some doubt the reality of this hypothetical RNA ALpha sequence, it is remarkable that among the millions of genetic sequences available to date, it is, curiously, the starfish genome which is closest to this sequence.

On the other hand, the demonstration of structures linked to the golden ratio and its alter ego the Fibonacci sequence are common here, in a fractal and self-similar manner simultaneously at the 2 nanometric scales of its DNA and macroscopic of the form geometric of the starfish.

Finally, in

(https://www.semanticscholar.org/paper/SIX-FRACTAL-CODES-OF-LIFE-FROM-BIOATOMS-ATOMIC-MASS-Perez-Montagnier/b332d3eed4a2f554c233da2e07ec68daab9a2182) we demonstrate (see figure) that, among the 3 biological universes DNA , RNA, proteins, the double strand of RNA, as we encounter in the transfer RNA of the universal genetic code, or here, in ALpha, Double stranded RNA constitutes a sort of zero or neutral element, unlike DNA and its proteomic images which are strongly correlated to it in what we call MASTER CODE of Biology (figure). This kind of zero could result physically, chemically and biologically in a great stability of these double-stranded RNAs for 2 reasons: this zero on the one hand, and the Fibonacci structures on the other hand.

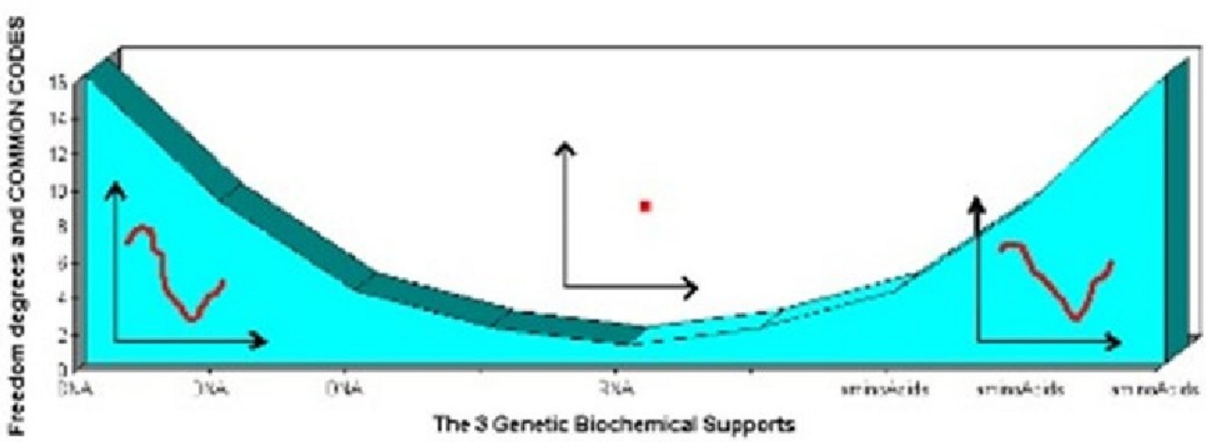
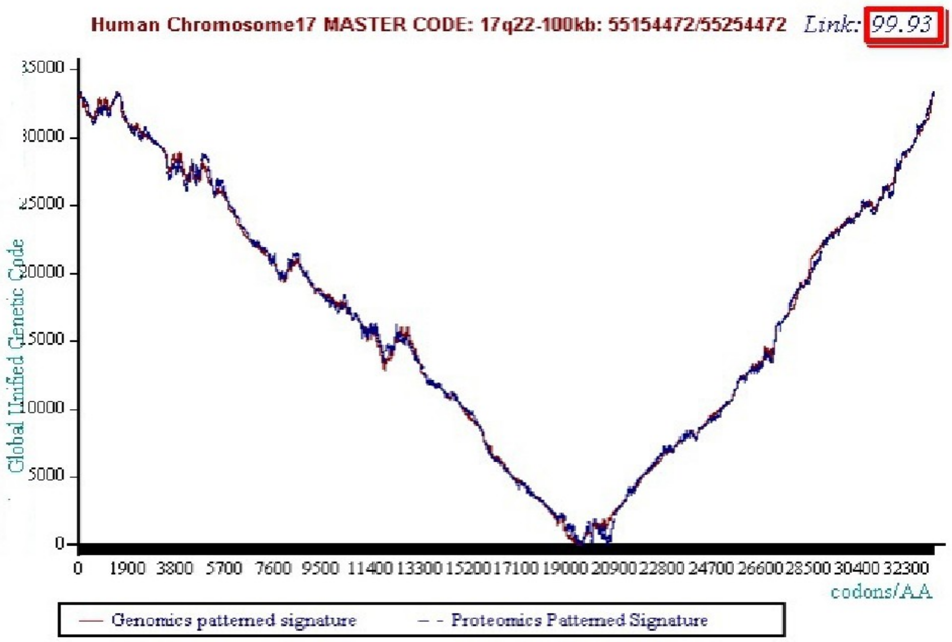


Figure 4 A symbolic representation of the 3 worlds of double stranded DNA (Genomics) hig corelated with potential double stranded amino acids (Proteomics) while RNA double stran

The RNA double strand constitutes a kind of ZERO of the Biology



The Master Code of Biology Unifying DNA and Proteomics Worlds

References :

(Perez, 1988), Perez J.C., De nouvelles voies vers l'Intelligence Artificielle, 1988, Ed. MASSON ELSEVIER, EAN 978-2225818158 ISBN 2225818150, https://livre.fnac.com/a223887/Jean-Claude-Perez-De-Nouvelles-voies-vers-l-intelligence-artificielle

(Perez, 1991), J.C. Perez (1991), "Chaos DNA and Neuro-computers: A Golden Link", in Speculations in Science and Technology vol. 14 no. 4, ISSN 0155-7785, January 1991
Speculations in Science and Cell Motility 14(4):155-7785
https://www.researchgate.net/publication/258439719_JC_Perez_1991_Chaos_DNA_and_Neuro-computers_A_Golden_Link_in_Speculations_in_Science_and_Technologyvol_14_no_4_ISSN_0155-778

(Perez, 1997), Perez J.C, L'ADN décrypté, Ed. Marco Pietteur, ISBN: 2-87211-017-8
EAN: 9782872110179, https://www.editionsmarcopietteur.com/resurgence/91-adn-decrypte-9782872110179.html

(Perez, 2009), Perez J.C, Codex biogenesis – Les 13 codes de l'ADN (French Edition) [Jean -Claude ... 2009);Language: French; ISBN -10: 2874340448; ISBN -13: 978-2874340444 https://www.amazon.fr/Codex-Biogenesis-13-codes-lADN/dp/2874340448

(Rapoport § Perez, 2018), Rapoport D. § Perez J.C, Golden ratio and Klein bottle Logophysics: The Keys of the Codes of Life and Cognition. Quantum Biosystems. 9(2) 8-76.; Vol. 9 – n.2 – 2018

(Varela § Maturana, 1980), Maturana H. & Varela F.J. (1980). Autopoiesis and cognition: the realization of the living. Reidel, Boston.

Demongeot J, Seligmann H. The Uroboros Theory of Life's Origin: 22-Nucleotide Theoretical Minimal RNA Rings Reflect Evolution of Genetic Code and tRNA-rRNA Translation Machineries. Acta Biotheor. 2019 Dec;67(4):273-297. doi: 10.1007/s10441-019-09356-w. Epub 2019 Aug 6. PMID: 31388859

Demongeot J, Thellier M. Primitive Oligomeric RNAs at the Origins of Life on Earth. Int J Mol Sci. 2023 Jan 23;24(3):2274. doi: 10.3390/ijms24032274. PMID: 36768599; PMCID: PMC9916791

(Perez, 2013) J-C perez, 2013, The "3 Genomic Numbers" Discovery: How Our Genome Single-Stranded DNA Sequence Is "Self-Designed" as a Numerical Whole, https://www.scirp.org/html/4-7401586_37457.htm